



Web Harvest RAG

A config-driven RAG chatbot starter — scrape any site or PDF, chunk and embed it, retrieve with a hybrid vector + keyword search, and chat over it. Built, benchmarked against a public IR baseline, and deployed to production.

Next.js 15

FastAPI

Supabase pgvector

FAISS

BM25

OpenAI

Docker + nginx

LIVE DEMO

webrag.huxiaoheng.com

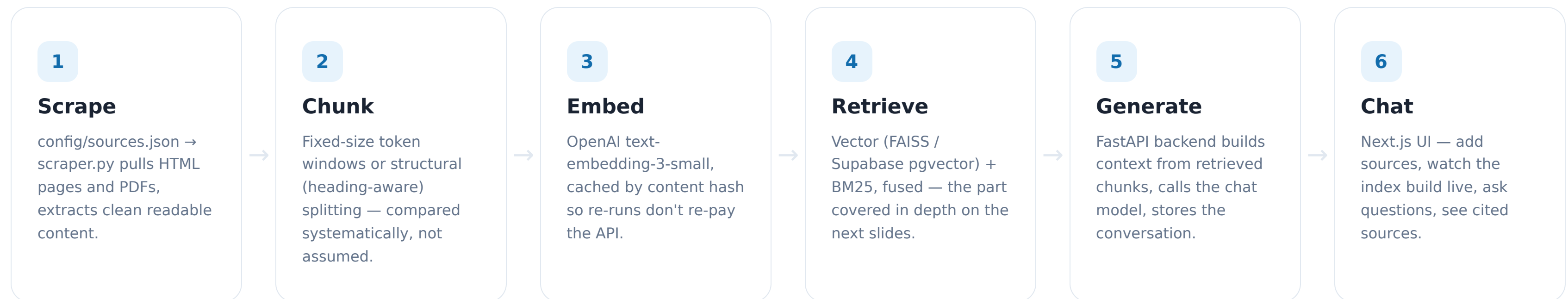
SOURCE

github.com/huxiaoheng44/web-harvest-rag

ARCHITECTURE

Scrape → Chunk → Embed → Retrieve → Chat

Six stages, each independently swappable — the retrieval stage is where most of the engineering work in this project actually happened.



Production uses **Supabase pgvector** for retrieval. A separate, standalone module — **retrieval_lab** — exists purely to answer "is this actually the best way to do stage 4?" without touching the live app. That's the rest of this deck.

retrieval_lab: a standalone experimentation harness

Ten questions about retrieval quality, each answered with real numbers instead of assumptions — kept fully separate from the production app.

1 Chunking strategies Fixed-size token windows vs. structural (heading-aware) splitting, compared head to head.	2 Vector vs. BM25 vs. Hybrid FAISS cosine search, rank_bm25 keyword search, and two fusion strategies (linear + RRF).
3 Cross-encoder rerank ms-marco-MiniLM reranking the hybrid candidate pool — tested, not assumed to help.	4 Chunk-size sweep 200 / 500 / 1,000-token chunks, measuring the completeness-vs-noise trade-off directly.
5 LLM query rewriting Paraphrase each query into variants, retrieve for each, merge via reciprocal rank fusion.	6 Error-bucket analysis Every imperfect-recall query auto-sorted into a failure category, then read through by hand.
7 Alpha tuning Hybrid fusion weight swept on a held-out validation split, evaluated on a separate test split.	8 LLM-as-judge answer eval Not just "was the right chunk retrieved" — is the generated answer actually faithful to it?
CORPUS A — MULTIVAC 24 scraped company pages/PDFs, 30 hand-labeled queries Web Harvest RAG	CORPUS B — NFCORPUS Public BEIR benchmark: 3,633 docs, 323 official test queries

Five real bugs, found by actually testing

Every one of these was caught by running the system end to end and looking hard at the output — not by code review alone.

SECURITY

Unauthenticated secrets endpoint
The in-app settings panel let anyone GET the real OpenAI/Supabase keys in plain JSON with zero auth. Disabled in production before the first public deploy.

CORRECTNESS

Zero-fill scoring bug in hybrid fusion
Candidates outside one method's top-20 were scored 0 instead of their real value — unfairly penalizing chunks only one method liked. Fixed with exact re-scoring over the full candidate union.

REPRODUCIBILITY

Non-deterministic recall@k
The same query, index, and alpha produced three different recall@1 scores across three runs — a bare Python set () 's hash-seed-dependent iteration order was silently reshuffling score ties.

EVAL CALIBRATION

LLM judge conflated two failure modes
A correct "I can't confirm this" refusal was scored as harshly as a hallucination. Average score jumped 3.32 → 4.92 / 5 once the judge prompt separated retrieval failure from generation failure.

PRODUCTION

Dependency drift broke the live chat
An unpinned supabase client picked up a newer API where .insert().select().single() no longer exists — every /chat call 500'd. Found by smoke-testing the fresh deploy, not by CI.

FINDINGS

What actually won, and where

MULTIVAC — structural-500

recall@k, 30 hand-labeled queries

METHOD	RECALL@1	RECALL@5
Vector only	60.0%	93.3%
BM25 only	75.0%	95.0%
Hybrid (linear)	78.3%	96.7%

NFCorpus — fixed-500 (BEIR)

recall@k, 323 official test queries, tuned $\alpha=0.8$

METHOD	RECALL@5	RECALL@100
Vector only	13.9%	35.4%
BM25 only	11.6%	23.0%
Hybrid (linear)	14.4%	36.0%

BM25 wins on MULTIVAC — brand-heavy content ("MLC", "PEAQ", "GS1 Digital Link") rewards exact keyword match. **Vector wins on NFCorpus** — technical medical abstracts vs. colloquial queries create a vocabulary gap that only semantic search bridges. Hybrid beats the better of the two on both corpora, but only once tuned per-corpus. Neither method is a safe default.

Validation: this harness's BM25 recall@100 on NFCorpus (23.0%) lands within 2 points of BEIR's published official BM25 baseline (25.0%) — the low absolute numbers are a property of the dataset, not a bug in the implementation.

LIVE

Deployed, not just demoed

webrag.huxiaoheng.com — a real conversation, retrieving from the production index.

W Web Harvest Chatbot

+ New chat

Add sources

Build status Idle

CURRENT SOURCES

No sources yet. Add a few URLs to build the knowledge base.

RECENT

What is MULTIVAC Line ... X

Portfolio Demo
website and PDF sources

What is MULTIVAC Line Control?
Web Harvest Chatbot - answers grounded in indexed sources

What is MULTIVAC Line Control?

WEB HARVEST CHATBOT

MULTIVAC Line Control (MLC) is a centralized solution designed for the operation and optimization of MULTIVAC packaging lines. It simplifies the management of recurring operating processes such as starting, stopping, and changing products, making these processes standardized and reproducible across the entire line. MLC allows for complete control from any terminal within the line, enhancing efficiency and reliability. It also includes features for line diagnostics, item management, and batch handling, contributing to improved quality and traceability of packs and products. The system supports integration with other production management systems like ERP, MES, MRP, and PPS for seamless operations.

Sources:

- MULTIVAC Line Control

SOURCES

MULTIVAC Line Control
https://multivac.com/int/en/our-solutions/portfolio/digital-solutions/multivac-line-control

MULTIVAC Line Control
https://multivac.com/int/en/our-solutions/portfolio/digital-solutions/multivac-line-control

Automation solutions
https://multivac.com/int/en/our-solutions/portfolio/automation-solutions

The MULTIVAC PEAQ Promise
https://multivac.com/content/dam/celum/04-Others/The-MULTIVAC-PEAQ-Promise/BR_PEAQPromise_EN_A4_144dpi_v2-0.pdf...

Digital solutions
https://multivac.com/int/en/our-solutions/portfolio/digital-solutions

Digital solutions
https://multivac.com/us/en/our-solutions/portfolio/digital-solutions

Ask about the indexed websites or PDFs...

Send

INFRASTRUCTURE

Docker Compose — isolated FastAPI + Next.js containers, bound to localhost only, restart: unless-stopped

EDGE

nginx reverse proxy + Let's Encrypt TLS, sharing the host with two other live projects

DATA

Production Supabase index rebuilt end to end: 81 → 340 chunks across all 24 source documents

VERIFIED

Real question in, real cited answer out — checked through the public HTTPS domain, not just localhost

Web Harvest RAG

06

STACK

Everything that shipped

Next.js 15

Chat UI, App Router

FastAPI

RAG, sources, conversations

Supabase

pgvector, match_chunks RPC

OpenAI

gpt-4o-mini + text-embedding-3-small

FAISS

Local vector search (retrieval_lab)

rank_bm25

Keyword search (retrieval_lab)

sentence-transformers

Cross-encoder rerank

Docker + nginx

Production deployment

LIVE DEMO

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